Online Appendix for "Timing Search"[1]

A Theory and model foundations

A.1 Illustrative foundation for log-quadratic profits

This subsection gives one illustrative construction of the main text's log-quadratic benchmark, $m(n) = \exp[-\frac{(1-\gamma)\varphi}{2}(n-n_p)^2]$. It is not a decentralized matching market and does not identify φ empirically.

Consider a market with n active firms. Each potential match between a firm and a trading partner has a quality θ drawn from a normal distribution $\theta \sim N\big(n_p, 1/[(1-\gamma)\varphi]\big)$. A firm operating at market thickness n captures matches with quality near n: specifically, matches within distance η of n on the quality dimension. The density of good matches available to the firm is proportional to the normal density evaluated at n:

$$m(n) \propto \sqrt{\frac{(1-\gamma)\varphi}{2\pi}} \exp\left(-\frac{(1-\gamma)\varphi}{2}(n-n_p)^2\right).$$

This adapts the spatial-competition logic of Hotelling (1929) and Salop (1979) to market thickness: firms compete for matches along a continuum (the quality-density dimension) rather than along a geographic line or circle, and the normal distribution generates the log-quadratic kernel. The connection to the equal-value structure of timing search runs through Burdett and Mortensen (1998), who carried the spatial-equilibrium logic into wage posting; the present paper carries the same logic into the time dimension via the timing-indifference condition. Absorbing the normalizing constant and applying the labor-reoptimization exponent $1/(1-\gamma)$ yields $\pi(n) = \pi_0 \exp[-\frac{\varphi}{2}(n-n_p)^2]$.

The parameter φ inherits its interpretation from the variance of match qualities: the curvature of $\ln m$ is $1/\mathrm{Var}(\theta)$, and labor reoptimization scales it into the profit curvature, $\varphi = 1/[(1-\gamma)\mathrm{Var}(\theta)]$. Industries with homogeneous match qualities (low variance, high φ) have a narrow peak and are sensitive to scale deviations. Industries with heterogeneous qualities (high variance, low φ) have a broad peak and are robust to deviations.

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A.2 Reduced-form timing friction

This appendix motivates the log-quadratic timing friction $\xi(\dot{n}) = e^{-\frac{\mu}{2}\dot{n}^2}$ as a convenient proportional cost of rapid *net adjustment*. It is not derived from gross entry market clearing.

Discrete-time formulation Consider a discrete-time economy with period length $\Delta > 0$. In each period, effective profit is reduced by a fraction proportional to the square of the adjustment *rate*:

$$y_t = \pi(n_t) \cdot \left[1 - \tfrac{\mu}{2}\left(\frac{n_{t+\Delta} - n_t}{\Delta}\right)^2\right].$$

The penalty depends on the rate $(\Delta n/\Delta)$, not the step: congestion comes from the speed of flow through the pipeline, so the fraction lost is invariant to the accounting period (a penalty scaled by Δ would vanish as $\Delta \to 0$ and support no friction in continuous time).

Continuous time and the canonical form As $\Delta \to 0$ with $\dot{n} = \lim_{\Delta \to 0}(n_{t+\Delta} - n_t)/\Delta$, the flow payoff is $\pi(n)\big[1 - \frac{\mu}{2}\dot{n}^2\big]$, which agrees with

$$y_t = \pi(n_t) \cdot e^{-\frac{\mu}{2}\dot{n}_t^2} = \pi(n_t) \cdot \xi(\dot{n}_t)$$

to second order in $\dot{n}$. The exponential is the canonical smooth representative (symmetric, one at $\dot{n} = 0$, $\ln \xi$ exactly quadratic, positive at all speeds). It has the required second-order behavior but is not a limit of the discrete penalty; the particular smooth form matters only beyond leading order (next paragraph).

Local quadratic approximation Any smooth symmetric friction $\xi(\dot{n})$ with $\xi(0) = 1$ and $\xi'(0) = 0$ admits the expansion $\ln \xi(\dot{n}) = -\frac{\mu}{2}\dot{n}^2 + O(\dot{n}^4)$, where $\mu = -\xi''(0)$. In a nonstationary classical full-support equilibrium, the leading-order frequency depends only on this local curvature. Symmetry and the use of net rather than gross flows remain substantive assumptions; the approximation does not establish that asymmetric or gross-flow models rotate.

A.3 Matching-motivated calibration of μ

This appendix details an illustrative calibration for Assumption 3. The main model treats μ as reduced form and does not impose the matching equation below as an equilibrium market-clearing condition.

Calibration thought experiment Consider $\mathcal{M}(S, R) = S^\alpha R^{1-\alpha}$, where S is a hypothetical participant mass and R indexes entry resources. The matching probability per participant is $p(\theta_e) = \theta_e^{-(1-\alpha)}$ with $\theta_e = S/R$. For units, normalize R so that reference throughput is $e_{ss} = \delta n_p$. This normalization does not imply that realized model entry satisfies $e_t = \mathcal{M}(S_t, R_t)$.

Steady state At $\dot{n} = 0$, the realized flow equals $e_{ss} = \delta n_p$, the infrastructure operates at its tuned capacity, and the matching probability equals $p_{ss} = p(\theta_e^{ss})$.

Bottleneck strain Gross entry is $e_t = \dot{n}_t + \delta n_t$ with reference flow $e_{ss} = \delta n_p$. The reduced form assumes that the slow replacement component is accommodated and that net adjustment pressure $\dot{n}_t$ loads a common adjustment pipeline: expansion strains permitting and construction, while contraction strains resale, scrapping, and reorganization. This symmetric treatment is imposed, not obtained from entrant matching. It also misses simultaneous large gross entry and exit when net flow is small.

I represent this strain by a CES-like penalty that enters multiplicatively into the matching probability:

$$p(\theta_e;\, \dot{n}) = \theta_e^{-(1-\alpha)} \cdot \left[1 + \frac{\dot{n}^2}{2(\delta n_p)^2}\right]^{-(1-\alpha)/\alpha}. \tag{1}$$

The strain term depends on $\dot{n}^2$ rather than on signed $\dot{n}$, encoding the symmetry between entry and exit congestion. The $(\delta n_p)^2$ scale in the denominator reflects infrastructure capacity calibrated to steady-state throughput.

Assumed versus derived The CES form in (1) is a modeling choice: any smooth strain index $g(x)$ entering as $[1+g(x)]^{-(1-\alpha)/\alpha}$, with $x = \dot{n}^2/(\delta n_p)^2$, $g(0) = 0$, and $g'(0) = 1/2$, gives the same second-order behavior (Online Appendix A.2). What it delivers is not the quadratic shape (universal) but the scaling $\mu = (1-\alpha)/[\alpha(\delta n_p)^2]$: $\mu \propto (\delta n_p)^{-2}$ by dimensional reasoning (the only scale for $\dot{n}$ is steady-state throughput δn_p), and in the $(1-\alpha)/\alpha$ factor the $(1-\alpha)$ comes from the Cobb–Douglas elasticity while the $1/\alpha$ is imposed by the selected CES strain normalization in (1); the scaling is a matching-motivated calibration, not a derivation, and is independent of the smoothing chosen.

Effective friction At fixed reference tightness, the illustrative strain ratio is $\ln \xi(\dot{n}) = -\frac{1-\alpha}{\alpha} \ln[1+\dot{n}^2/(2(\delta n_p)^2)]$. For small $|\dot{n}|/(\delta n_p)$ this gives $\ln \xi(\dot{n}) \approx -\frac{\mu}{2}\dot{n}^2$ with $\mu = \frac{1-\alpha}{\alpha(\delta n_p)^2} > 0$. This supplies a scale for μ, not an identified structural mapping. All main results use $\mu > 0$ directly.

B Supporting theory and perturbations

B.1 Conditional spectral implication

This appendix records the leading-order calculation behind the slow perturbation mentioned in the main text. It starts from the harmonic cycle of Theorem 1(ii) of the main text and imposes smooth phase-continuing modulation as the payoff locus moves. Because the locus now changes over time, the deterministic classification does not apply directly. The calculation conditions on phase continuation and does not claim a stochastic equilibrium.

Competitive participation on active dates gives $V_t = \kappa_t$, so asset pricing implies

$$\pi(n_t)\xi(\dot{n}_t) = (r+\delta)\kappa_t - \frac{\mathbb{E}_t[d\kappa_t]}{dt}. \tag{2}$$

For a small, slowly moving log-cost perturbation, the associated payoff shortfall is, to first order, an affine mean-reverting process. Writing its effective coefficients as $(\bar{b}, \theta, \sigma_\kappa)$,

$$db_t = -\theta(b_t - \bar{b})\,dt - \sigma_\kappa\,dW_t.$$

The raw Ornstein–Uhlenbeck process is used only as a moment reference. Because it is unbounded and nowhere differentiable, the economic construction instead requires a bounded smooth process with matching leading moments.

If $\sigma_\kappa/\sqrt{2\theta} \ll \bar{b}$ and $\theta \ll \omega_0$, freezing b_t over one turn gives the conditional representation

$$n_t - n_p \approx \sqrt{\frac{2b_t}{\varphi}}\cos(\omega_0 t + \phi_0), \qquad \omega_0 = \sqrt{\varphi/\mu}.$$

Thus the imposed phase law gives the candidate power near the nonzero angular frequency ω_0, with slow amplitude variation spreading spectral mass into a band around ω_0.

B.2 Conditional amplitude moments

The reference shortfall law is

$$db_t = -\theta(b_t - \bar{b})\,dt - \sigma_\kappa\,dW_t, \qquad b_t \sim \mathcal{N}\left(\bar{b}, \frac{\sigma_\kappa^2}{2\theta}\right) \quad \text{in stationarity.} \tag{3}$$

A Gaussian shortfall can be nonpositive, in which case the square-root amplitude is undefined. The formulas below therefore require that this probability be negligible, or that the bounded smooth economic process be explicitly truncated away from zero.

Averaging the phase of the phase-continuing candidate gives

$$\mathbb{E}[(n_t - n_p)^2] \approx \frac{\bar{b}}{\varphi}, \qquad \mathbb{E}[\dot{n}_t^2] \approx \frac{\bar{b}}{\mu}, \qquad \mathbb{E}[(n_t - n_p)\dot{n}_t] \approx 0. \tag{4}$$

These are conditional moments of a quasi-static construction. The identity

$$\frac{\varphi}{2}\mathbb{E}[(n_t - n_p)^2] + \frac{\mu}{2}\mathbb{E}[\dot{n}_t^2] \approx \bar{b}$$

is a per-establishment effective-profit accounting relation, not household welfare.

B.3 Additional properties of the general period

Let $h(n) = \ln[\pi(n)/\bar{y}]$. The two one-way travel times from the profit peak to the break-even scales are

$$T_- = \sqrt{\frac{\mu}{2}}\int_{n_-}^{n_p} \frac{\mathrm{d}n}{\sqrt{h(n)}}, \qquad T_+ = \sqrt{\frac{\mu}{2}}\int_{n_p}^{n_+} \frac{\mathrm{d}n}{\sqrt{h(n)}}.$$

The full period of the general-profit orbit (Section 3.1 of the main text) is $2(T_- + T_+)$. Thus asymmetry in log profit has a direct timing interpretation: it changes the time spent on the two sides of the profit peak even though the same payoff level determines both turning points.

For completeness, impose the additional local-curvature condition $\pi''(n_p) < 0$, and let $\varepsilon = \ln[\pi(n_p)/\bar{y}]$ and $k = -\pi''(n_p)/\pi(n_p) > 0$. As $\varepsilon \downarrow 0$,

$$h(n_p + x) = \varepsilon - \frac{k}{2}x^2 + o(x^2), \qquad n_\pm - n_p = \pm\sqrt{\frac{2\varepsilon}{k}} + o(\sqrt{\varepsilon}).$$

Changing variables by $x = \sqrt{2\varepsilon/k}\,u$ in the general period integral gives

$$T(\bar{y}) \longrightarrow 2\pi\sqrt{\frac{\mu}{k}}, \qquad \frac{2\pi}{T(\bar{y})} \longrightarrow \sqrt{\frac{-\pi''(n_p)}{\mu\pi(n_p)}}.$$

This is the local limit noted in Section 3.1 of the main text. Higher-order smoothness is needed only to calculate finite-amplitude corrections, not for the local limit or the global classification within the classical full-support class.

Isochrony and the width of the well The period depends on the profit landscape only through the width of the well. Write $w(n) \equiv \ln[\pi(n_p)/\pi(n)]$ for the log-profit shortfall and $\ell(w) \equiv n_+(w) - n_-(w)$ for the distance between the two scales at shortfall w; a change of variables in the period integral gives the half-period at level u as $\sqrt{\mu/2}\int_0^u \ell'(w)\,(u-w)^{-1/2}\,\mathrm{d}w$, which involves the landscape only through ℓ. Two landscapes with the same width function therefore share every period. The log-quadratic landscape has $\ell(w) = 2\sqrt{2w/\varphi}$, so any landscape with this width function, however skewed, is isochronous at the same $\omega = \sqrt{\varphi/\mu}$, with φ read as the curvature of the width function. Prediction 2's estimand therefore tolerates landscape skewness, consistent with the time-reversal equality of the expansion and contraction legs in Online Appendix B.5.

B.4 Entry at all dates: full-support feasibility

This appendix verifies positive gross entry and a slack outside pool for the harmonic candidate paths in Theorem 1(ii) of the main text.

Proposition 6 (Full-support feasibility bound).
The harmonic candidate paths in Theorem 1(ii) of the main text satisfy full-support feasibility (a slack outside pool $M - n_t > 0$ and a positive entry flow $e_t = \dot{n}_t + \delta n_t > 0$ at every date) if and only if the cycle amplitude satisfies the strict joint upper bound

$$A \;<\; \min\left\{ \frac{\delta n_p}{\sqrt{\omega^2 + \delta^2}}\,,\; M - n_p \right\}. \tag{5}$$

The first term is the entry-flow bound: the steady-state exit flow δn_p must exceed the maximum amplitude of net-flow oscillation. The second is the pool bound: the active population must remain strictly below M.

Proof. Along the candidate path, $n_t = n_p + A\cos(\omega t + \phi)$, so the outside pool $M - n_t$ attains its minimum $M - n_p - A$ (the cosine attains 1), and it is positive at every date iff $A < M - n_p$, the second term of (5).

The entry flow is $e_t = \dot{n}_t + \delta n_t = -A\omega \sin(\omega t + \phi) + \delta(n_p + A\cos(\omega t + \phi))$. The non-constant component has maximum modulus $A\sqrt{\omega^2 + \delta^2}$, so $e_t > 0$ whenever $A < \delta n_p / \sqrt{\omega^2 + \delta^2}$. In the conditional stochastic illustration, exact positivity also requires slack for the slowly moving amplitude; the deterministic bound then applies only to leading order.

The strict joint condition $A < \min\{\cdot, \cdot\}$ is therefore necessary and sufficient. ∎

Empirical scale diagnostic The bound applies to the establishment count, the model's state. In the earlier 15-industry sample, I approximate A/n_p by $\sqrt{2}$ times the HP-filtered standard deviation of annual log BDS establishment counts and pair it with each industry's current fixed-band production frequency. The exact bound holds for 14 of 15 industries at both $\delta = 0.065$ and 0.090. Across the observed frequency range $[0.99, 3.06]$ radians per year and those exit rates, the bound permits amplitudes of roughly 2–9% of scale. This is a hybrid scale diagnostic, not a structural test: it combines count amplitudes with production frequencies. It confirms that the full-support harmonic equilibrium is a small-amplitude benchmark; applying its ranking to larger production movements is an extrapolation.

B.5 Frequency correction and time across scales

The exact-harmonic result of Theorem 1(ii) of the main text is specific to the log-quadratic profile. This subsection records how the frequency and the cycle shape respond when the profile is non-log-quadratic; the statements are confirmed numerically in Online Appendix C.

Finite-amplitude frequency. For a smooth profit function with a strict peak at n_p, the small-amplitude angular frequency in a nonstationary classical full-support equilibrium is

$$\omega_0 = \sqrt{\frac{|\pi''(n_p)|}{\mu\,\pi(n_p)}}.$$

At finite amplitude it generally acquires an $O(A^2)$ correction. The log-quadratic profile is the special amplitude-independent case. Across eight smooth single-peaked specifications the corresponding period stays within 1.6% of the numerical rotating-path period at $A/n_p = 0.30$.

Time across scales. When the profit landscape is skewed ($\pi'''(n_p) \neq 0$), the orbit can spend unequal time above and below n_p. Because $|\dot{n}|$ depends only on the current scale, however, the expansion and contraction legs are exact time reversals and each lasts one-half of the full cycle. Landscape skewness therefore changes time across scale regions, not the duration of expansion relative to contraction. The simulations confirm this distinction (Online Appendix C).

B.6 Equilibrium verification

This appendix verifies that the harmonic paths in Theorem 1(ii) of the main text satisfy all four conditions of the full-support timing-indifference equilibrium (Definition 1 of the main text).

Entry policy and aggregate consistency (iii) The entry policy implied by the equilibrium path is $e_t = \dot{n}_t + \delta n_t$. Using $n_t = n_p + A\cos(\omega t + \phi)$:

$$e_t = -A\omega \sin(\omega t + \phi) + \delta(n_p + A\cos(\omega t + \phi)).$$

This is continuous and satisfies $\dot{n} = e - \delta n$ by construction.

Optimality and participation (i, ii) The flow free-entry condition gives $\pi(n_t)\xi(\dot{n}_t) = C = (r+\delta)\kappa$ for all t, so $V(\tau) = C/(r+\delta) = \kappa$ for every τ. Every active date is therefore optimal and no participant gains by retiming. With a slack outside pool, competitive participation gives zero waiting surplus. Entry non-negativity holds under the amplitude bound in Online Appendix B.4.

Feasibility (iv) The maximum of n_t is $n_p + A$. Feasibility requires $n_p + A < M$, so the outside pool remains slack on the harmonic path.

C Numerical robustness: beyond the log-quadratic

This appendix checks the log-quadratic frequency formula against non-log-quadratic profiles. Across a library of 8 smooth exponential-family single-peaked specifications (log-quadratic at $\varphi \in \{1, 4, 9\}$ with quartic and cubic perturbations), the worst-case error from using $T_0 = 2\pi\sqrt{\mu/\varphi}$ in place of the exact full-cycle period rises from 0.05% at $A/n_p = 0.05$ through 0.74% at 0.20 to 1.64% at 0.30.

Conditional rotating-path simulations illustrate two comparative statics. Quartic profit terms generate the predicted finite-amplitude frequency correction, while an asymmetric profit landscape changes the share of time spent above and below the peak. Expansion and contraction remain time reversals. These are numerical properties of the imposed traversal.

D Empirical detail

The main paper reports the fixed-protocol concentration ranking, the annual and quarterly establishment checks, the depth–recovery diagnostic, and two historical paths. This section supplies construction details and archives broader exploratory exercises. None independently measures profit curvature or the timing friction.

D.1 Concentration index and fixed-protocol universe

The ranking uses an aggregate FRED industrial-production series for each of the 21 three-digit manufacturing categories. Each ID is `IPGxxxS`, with `xxx` equal to the NAICS-3 code; the pinned production IDs and full sample accounting are in `data/sample_universe_data.json`.

The regressor is the Ellison and Glaeser (1997) index

$$\gamma \;=\; \frac{G - \left(1 - \sum_i x_i^2\right) H}{\left(1 - \sum_i x_i^2\right)(1 - H)}, \qquad G = \sum_i (s_i - x_i)^2,$$

where s_i is area i's share of the industry's employment, x_i its share of total manufacturing employment, and H the plant-size Herfindahl. Every input is public: G comes from the County Business Patterns county or state file, and H from the national file, whose employment and establishment counts by size class give each class's actual mean plant size. The index is computed at both geographies for 2016 and 2021. The baseline is the state level in 2021, the geography of Ellison and Glaeser (1997) and the latest cached year, and every regression is reported across the four variants. The construction is `codes/build_eg_concentration.py`; the regressions are `codes/eg_built_regressions.py`, with output `data/eg_built_regressions.json`.

Legacy archived score Earlier drafts used a fixed, rounded score vector stored in `codes/ellison_glaeser.py`; its underlying observations, SIC–NAICS concordance, aggregation weights, and construction program are not archived, so it is neither a published nor an independently reproducible series. In logs it correlates 0.73–0.81 with the built index across the 21 industries. It is retained only for the legacy 15-industry exercises below; under it the production slope is 0.16 (HC1 0.12) and the BDS establishment slope is 0.29 (HC1 0.11), against 0.06 (HC1 0.08) and 0.18 (HC1 0.09) under the built index.

NAICS	γ	NAICS	γ	NAICS	γ
311	.009	312	.040	313	.051
314	.013	315	.078	316	.077
321	.015	322	.017	323	.004
324	.092	325	.042	326	.006
327	.011	331	.049	332	.004
333	.012	334	.064	335	.018
336	.034	337	.006	339	.020

These are the complete legacy scores, not verified published estimates.

D.2 Depth and recovery time: testing the equilibrium response

The equilibrium response of Proposition 3 maps recessions into displacements. A transient shock displaces the scale from a cycle peak by Δ, and the dynamically robust continuation re-enters the same orbit, so recovery to the prior peak takes $\arccos(1-\Delta/A)/\omega$: log recovery time on log depth has slope one-half at leading order, and between 0.51 and 0.55 for the observed depth distribution when the calibrated amplitude keeps troughs within the upper half of the orbit; the local slope rises toward one as a displacement approaches the full swing. Direction is free in Proposition 3, and the design is direction-proof: the measured trough is the realized minimum of the path, so a continuation that falls first registers the full swing as its depth and the half period as its recovery, the endpoint of the same curve. Mapping depth into orbit amplitude instead predicts slope zero. Recovery that closes the gap at a roughly constant rate, as trend growth does, gives duration proportional to depth, slope one; pure exponential reversion toward the prior peak never crosses it in finite time, and against a fixed threshold it gives a nearly flat logarithmic relation, which the rejection of slope zero also rejects.

I examine the prediction in monthly production for 21 NAICS-3 manufacturing industries across seven NBER recessions from 1972 through 2024. For each industry–recession cell, I locate the industry's own peak and trough within nine months of the NBER dates, retain declines exceeding two log points, and define recovery as the first month production regains its pre-recession peak, right-censored after 60 months. There are 134 qualifying episodes: 85 recover and 49 are censored.

In the recovering sample, log recovery months on log depth, with industry fixed effects and standard errors clustered by recession, gives 0.469 (s.e. 0.179): the zero of the amplitude mapping is rejected ($t = 2.62$; t_6 reference $p = 0.040$), the constant-rate one is rejected ($t = -2.97$), and one-half is not ($t = -0.17$). Leave-one-recession-out slopes span 0.31 to 0.58. The phase mapping presumes a recoverable prior peak, so the clean subsample is the fifteen industries with positive full-sample production growth: there the slope is 0.492 (s.e. 0.190, $N = 67$), with both alternatives still rejected at the five-percent level on a t_6 reference.

Censoring is consistent with the non-recovering branch. Censored episodes are deeper (median log depth 0.22 against 0.15; Mann–Whitney $p < 10^{-5}$), industry trend growth predicts censoring given depth (linear-probability coefficient -3.7, s.e. 0.8), and censoring is 54 percent in declining industries against 29 percent in growing ones: the signature of permanent downshifts and withering, for which the prior peak is no longer the target, rather than of slow recoveries. An accelerated-failure-time regression that instead pools all 134 episodes as one censored process returns 0.96, exponent one, which is what conflating the regimes produces. Log peak-to-trough duration on log depth gives 0.146 (s.e. 0.177): the decline is the displacement, not orbit traversal. Recovery times also rise with the industry's own spectral period (0.19, s.e. 0.13); unit scaling attenuated by the peak-estimation noise of the main text's Section 4.1 (reliability about 0.25) reconciles the coefficient, a consistency check rather than a sharp test.

Production is not the model's establishment state and recession dating is inherited from the NBER; seven clusters bound the precision throughout. The design is implemented in `codes/isochrony_panel.py` and `codes/isochrony_regime.py`; episode-level output is in `data/isochrony_panel_data.json` and `data/isochrony_regime_data.json`.

D.3 Historical echo: manufacturing and oil investment

This appendix details a descriptive historical diagnostic. BEA investment in structures bundles establishment creation with capacity expansion, so vintage replacement (Kydland and Prescott, 1982), structural breaks, and other propagation mechanisms are important confounds. Within each chosen window, I demean annual investment growth, evaluate the raw periodogram at the positive Fourier frequencies, select its largest ordinate, and fit a constant, cosine, and sine at that frequency by OLS. Fisher's g is the largest ordinate divided by their sum. Conditional on a fixed series and window, its white-noise reference distribution accounts for taking the maximum over the Fourier grid; it does not account for the post-hoc choice of industries, windows, transformation, or displayed specification.

The fitted event-window components are 5.0 years for manufacturing in 1934–1973 and 3.25 years for mining exploration in 1979–2004. Their nominal Fisher-g p-values are 0.045 and 0.026. The equal-length comparison-window values are 0.637 for manufacturing and 0.027 for mining, whose pre-1979 peak is the full 26-year window. The white-noise reference is itself fragile: after fitting the event-window cosine, residual ACF(1) is 0.307 for manufacturing (a simple lag-one Ljung–Box $p = 0.044$) and 0.252 for mining ($p = 0.174$). The industries and windows were chosen after inspection, so the main text omits these p-values and uses the plots only to make differing rhythms visible. The construction is archived in `codes/echo_analysis.py` and `data/echo_analysis_data.json`.

D.4 External proxies and the historical-IV exercise

In the log-quadratic benchmark, conditional on a classical full-support cyclical equilibrium, the coefficients on the unobserved primitives $\ln \varphi_i$ and $\ln \mu_i$ are $1/2$ and $-1/2$. The implemented regression instead uses the built Ellison–Glaeser index γ_i. Its slope equals one-half only if concentration maps one-for-one into curvature and the timing friction is orthogonal to concentration. Neither condition is identified here. The instrument exercises below predate the built index and use the legacy archived score.

Two available proxies for μ_i (the BEA capital replacement horizon and a plant-level adjustment coefficient) are wrong-signed and insignificant in frequency regressions. They mainly measure durability and capital intensity, so this failure is evidence that they do not isolate the model's pipeline friction, not evidence for μ-uniformity. Federal regulatory restrictions are positively related to concentration at the 10% level, which makes orthogonality less rather than more secure. Adding these controls to the restricted 15-industry regressions leaves the concentration coefficient broadly similar, but small samples and weak first stages prevent a clean two-primitive test.

The historical-geography instruments are likewise restricted to 15 hand-mapped industries. The broad 1947 score has a robust first-stage statistic of 17.81, but the conventional statistic is below the Stock–Yogo threshold. Its estimated slope is 0.531 under the inherited detection rule, 0.262 after dropping Printing's roughly 28-year peak, and 0.158 when every peak is re-estimated inside the fixed 2–12-year band. These estimates are reported as robustness to concentration endogeneity; the instrument does not establish that γ_i measures φ_i, nor does it extend to the full 21-industry universe.

Finally, the revealed-curvature calculation constructs curvature from the same estimated frequency used in the outcome. It is an internal algebra check, not independent evidence on the concentration-to-curvature elasticity. Direct measures of the profit landscape and permitting or construction delays are needed for a structural test.

D.5 Spectral robustness panel

This appendix reports an alternative-estimator panel for the earlier restricted $N = 15$ sample under the legacy archived score (Table D.1). The more mechanical comparison uses all 21 industries, five DPSS tapers, and a fixed 2–12-year band; both designs are exploratory appendix evidence.

Table D.1: Cross-industry slope under six spectral estimators

Estimator	$\hat{\beta}_\gamma$	HC1 SE	p-value	R^2
Burg AR(24) (higher order)	0.240	0.069	0.004	0.482
Welch periodogram (Hann)	0.309	0.099	0.008	0.333
Multitaper DPSS ($K = 5$)	0.497	0.140	0.004	0.556
Burg AR(12)	0.531	0.216	0.029	0.299
Yule-Walker, BIC order	0.137	0.390	0.731	0.010
Welch (long segments)	0.103	0.118	0.396	0.056

Notes: Restricted-sample OLS regressions of log estimated frequency on log concentration, with HC1 standard errors (White, 1980). The estimates span 0.10–0.53, showing material sensitivity to spectral method. No estimator is ranked by proximity to one-half: higher-order autoregressions may reveal genuine multimodality, while low-order and smoothed estimators make different bias–variance tradeoffs. The all-industry fixed-protocol estimate is reported below as the common-rule comparison.

AR(24) prewhitening robustness Prewhitening leaves the AR(24) slope in $[0.219, 0.238]$. AR(24) often selects a higher-frequency mode than multitaper or AR(12) in long-cycle industries. The exercise documents genuine estimator dependence; it is not grounds for discarding the lower estimate.

D.6 Simulation diagnostics and establishment-based peaks

HC1 inference treats each estimated peak as observed and industries as independent. The taper jackknife below measures peak-location instability. Most joint-bootstrap code, however, was written for the earlier restricted 15-industry design; it is reported as a robustness diagnostic and is not a confidence interval for the all-21 estimate. Implemented in the replication scripts under `codes/`.

Peak uncertainty The delete-one-taper jackknife gives a median SE[$\ln\hat{\omega}_i$] of 0.09, heterogeneous across industries (mid-band industries reach 0.3–0.9; Plastics 0.91). This is a within-sample precision measure and understates cross-sample sampling variation; the headline specificity test in the main text uses the parametric bootstrap standard deviation of 0.47 against the strict-band 21-industry dispersion of 0.54, which is the comparison that determines whether the spread exceeds estimation error; the corresponding common-frequency test is $\chi^2(20) = 26.83$, $p = 0.14$. The wide detection-band 15-industry dispersion is 0.71.

Restricted-sample joint bootstraps Two fitted panel processes re-estimate peaks and the earlier 15-industry OLS/IV slopes. Results vary sharply with the assumed data-generating process: a peak-preserving process re-centers slopes near zero, whereas a moving-block bootstrap puts most mass above zero. These diagnostics underscore estimator and DGP uncertainty; they do not rescue a coefficient of one-half or supply full-universe inference.

Establishment-based peaks Re-estimating each frequency from log-differenced BDS establishment counts (the model's state; 4-digit to NAICS-3, annual 1978–2022, Burg AR(6), 3 of 15 edge-censored) gives log-frequency peaks that correlate 0.32 with production peaks under the fixed 2–12-year production protocol (Pearson $p = 0.24$) and a concentration slope of +0.18 (HC1 0.09) under the built index, +0.29 (HC1 0.11) under the legacy score. A quarterly upgrade from QCEW establishment counts (national NAICS-3, 1990–2024) fails informatively: quarterly net entry is small relative to the stock, a common component absorbs a median 63% of growth variance, and the raw peaks cluster at the 9.5–11.9 year spacing of the 1991–2020 recessions. Neither the raw nor factor-adjusted frequencies yield a positive concentration ranking (slopes under the built index: OLS −0.012, factor-adjusted −0.044; legacy-score IV −0.003). This is a direct-state non-confirmation, not a prediction of the model. Production growth on the same 1990–2024 window retains the wide detection-band peak dispersion (sd $\ln\hat{\omega} = 0.71$; cf. the strict-band 0.54 in the main text) and a positive slope under the legacy score (+0.25 detection, +0.16 strict; below full sample, a stability caveat).

Built-index bootstrap and recession-timing purge Two checks discipline the headline cells under the built index (state, 2021). First, a parametric bootstrap refits each industry's dynamics (AR(12) on monthly production growth; AR(6) on annual establishment growth), simulates, re-estimates every peak by the design's own estimator, and re-runs the slope: production gives mean +0.04, sd 0.08, $P(\hat{\beta} > 0) = 0.70$; establishments give mean +0.12, sd 0.085, a 90% interval $[-0.02, 0.25]$, $P(\hat{\beta} > 0) = 0.91$, with the simulated band-edge count matching the data's three. The bootstrap standard deviations equal the HC1 standard errors, so the reported precision is not understated. Second, purging a cross-industry common factor or NBER recession dummies from monthly production growth before peak estimation moves the production slope from +0.06 to +0.08 and +0.07: the ranking is not a recession-harmonic artifact, although individual peaks do move (correlation 0.70 with the unpurged peaks). Scripts: `codes/production_purge_bootstrap.py` and `codes/bds_slope_bootstrap.py`.

Sample accounting All 21 NAICS-3 manufacturing industries have the required production series and a built index value. The fixed-protocol specification retains all 21 and applies the same 2–12-year multitaper rule, giving +0.06 (HC1 0.08) under the built index and +0.16 (HC1 0.12) under the legacy score. The earlier 15-industry sample omitted six industries through a mixture of documented boundary screens and inherited exclusions; it is retained only for replication and IV robustness. The historical instrument set is hand-mapped only for those 15 and does not extend to the full universe.

Reading The full-universe production association is positive but imprecise, and the annual establishment ranking is positive and distinguishable from zero under the built index, while quarterly establishments provide a direct-state non-confirmation. The evidence supports a qualitative model-inspired ranking; one-half remains conditional on unverified proxy and friction restrictions.

E The planner's problem

This appendix derives the planner's optimality condition, the flat cycle-eliminating implementation, and the state-dependent Pigouvian wedge of the main text, maintaining the paper's common discount rate $r = \rho$.

Planner's problem The planner chooses the entry flow $e_t \geq 0$ to maximize household effective consumption (consumption net of labor disutility and entry resources), which reduces to

$$\max_{\{e_t\}} \int_0^\infty e^{-\rho t}\big[\, n_t\,\pi(n_t)\,\xi(\dot{n}_t) - \kappa\, e_t \,\big]\,\mathrm{d}t, \qquad \dot{n}_t = e_t - \delta n_t,$$

where $n_t\pi(n_t)\xi(\dot{n}_t)$ is aggregate effective profit, the operating contribution to household effective consumption under the main text's quasilinear metric, and κe_t is the entry-resource cost. Substituting $e_t = \dot{n}_t + \delta n_t$, the integrand is $\mathcal{F} = e^{-\rho t}[\, n\pi(n)\xi(\dot{n}) - \kappa\dot{n} - \kappa\delta n\,]$, and the Euler–Lagrange condition $\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{F}_{\dot{n}} = \mathcal{F}_n$ reads

$$-\rho\big(n\pi\xi' - \kappa\big) + \big(\pi\xi' + n\pi'\xi'\big)\dot{n} + n\pi\xi''\,\ddot{n} \;=\; \pi\xi + n\pi'\xi - \kappa\delta.$$

At a steady state ($\dot{n} = \ddot{n} = 0$), $\xi(0) = 1$ and $\xi'(0) = 0$, so the left side is $\rho\kappa$ and the condition reduces to

$$\pi(n^*) + n^*\pi'(n^*) = (\rho + \delta)\,\kappa,$$

the marginal aggregate profit of an entrant equated to the entry-cost annuity. To leading order around n_p the objective is concave, so a small cycle lowers time-averaged effective consumption and the optimum is the rest point; the global no-cycle statement requires concavity of $n\pi(n)\xi(\dot{n})$, not merely of π (Jensen applied to the individual factors does not extend to their product), so no-cycling is a local, leading-order property. As the entry surplus vanishes ($(\rho + \delta)\kappa \to \pi(n_p)$), $\pi'(n_p) = 0$ gives $n^* \to n_p$.

Flat cycle-eliminating implementation A lump-sum-rebated entry wedge τ sets $V_t = \kappa + \tau$, so the flow free-entry condition becomes $\pi(n_t)\xi(\dot{n}_t) = (r + \delta)(\kappa + \tau)$. Choosing $\tau^{\text{collapse}} = \pi(n_p)/(r + \delta) - \kappa$ makes the full-support locus $\pi(n)\xi(\dot{n}) = \pi(n_p)$, which shrinks algebraically to $(n_p, 0)$. Under ordinary complementarity it also creates a no-entry equilibrium from any inherited n_0: $e_t = 0$ and $n_t = n_0 e^{-\delta t}$ imply $\pi(n_t)\xi(-\delta n_t) \leq \pi(n_p)$, so entry value is at most the taxed cost $\pi(n_p)/(r + \delta)$. Thus the flat wedge supports both parking at n_p and extinction from any inherited scale. It also does not price marginal harm, implement the planner's generally distinct n^*, or characterize a transition to n_p. A rebated wedge is a transfer, whereas raising the real entry cost consumes resources; no general welfare ranking follows from the locus calculation alone.

The state-dependent Pigouvian wedge Within the full-support class, the flat wedge leaves only n_p, not the planner's generally distinct rest point n^*; the wedge that decentralizes the planner's path prices both externalities state by state. Let τ_t be rebated lump-sum, so full-support participation reads $V_t = \kappa + \tau_t$.

Proposition 5 of the main text states the wedge and gives its derivation: the planner's Hamiltonian yields an entry margin and a costate equation whose forward solution, substituted into the wedge, collapses to $\tau_t = V_t - \kappa$, so that private free entry holds along the planner's path.

The first term is congestion pricing on entry timing ($\xi' = -\mu\dot{n}\,\xi$): a tax during expansions, a subsidy during contractions. The integral capitalizes the scale externality $-n\pi'$ the entrant imposes on incumbents, net of the exit-replacement interaction $\delta n\pi\xi'$. At a rest point the wedge is $\tau^* = -n^*\pi'(n^*)/(\rho + \delta)$, which is positive when the optimum lies on the crowded side of the peak, $n^* > n_p$; its sign is not unconditional. The private value then reproduces the planner condition. This state-dependent object is the Pigouvian wedge; the flat cycle-eliminating wedge is not.

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